Survival of the Fitted: Cross-Checkpoint Transfer of a Jacobian Lens from Qwen3.6-27B to Qwen3.8-27B

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Abstract

Jacobian lenses map intermediate language-model states into vocabulary-space readouts of verbalizable content [3], but they are usually fitted and evaluated on a single frozen checkpoint. Tuned-lens translators are known to transfer from a base model to its fine-tuned variants with a small next-token cost [2]. Whether a Jacobian lens preserves its latent semantic readouts and its causal directions across a model-version change has not been measured, so a change in a readout after an update cannot yet be separated from mismatch between the lens and its new target. We applied a lens fitted on Qwen3.6-27B unchanged to Qwen3.8-27B. We evaluated the transferred lens on two tasks: recovering an unstated country needed to connect two facts and ranking the next token in held-out WikiText-103 passages. We also tested whether lens-derived directions could causally alter generated wording. Transferring the fitted lens across this architecture-compatible model-version change weakened the latent-entity signal late in the network without erasing it: median target rank over 40 prompts rose from 4 to 17 at layer 48, while layers 24 to 36 favored the successor, with a median rank of 38 at layer 24 against 121 on the home model. The next-token readout showed a distinct and depth-dependent transfer cost: paired rank inflation of 1.2–1.3× at mid-network layers and about 2× by layer 48; the raw readout shifted by a similar 1.2–1.3× at mid-network layers, so the lens-specific cost is concentrated near layer 48. Across both checkpoints, with reasoning mode enabled and disabled, removing lens-derived directions for *paradox* reduced the corresponding readout to zero and eliminated the word from every ablated generation. Together, these results show that a fitted lens can remain informative across a nearby, architecture-compatible model-version change without preserving its original calibration. Interpretability and alignment evaluations that compare model versions should therefore measure transfer error before attributing readout differences to the model.

1 Introduction

The Jacobian lens is a method for interpreting intermediate states in a language model. For each layer, it estimates a linear transport from that layer into the final-layer representational basis. Applying the model's own unembedding after this transport produces a vocabulary-space readout. This readout is intended to recover verbalizable content that is difficult to see with the raw logit lens, which applies the unembedding directly to an intermediate state. Gurnee et al. [3] use the method to support a global-workspace account of verbalizable mid-network representations. The tuned lens provides a related vocabulary-space readout but learns a separate translator for each layer [2]; the raw logit lens uses no learned or geometry-derived transport [9].

These methods are normally fitted and evaluated on a single frozen checkpoint. The nearest test of reuse is the tuned lens: translators trained on LLaMA 13B transfer to Vicuna 13B, an instruction fine-tune of the same base, with at most a 0.3 bit-per-byte increase in divergence from the model's output [2]. That result covers a trained translator, a documented fine-tune of a fixed base, and a next-token fidelity metric. It does not say whether a geometry-derived transport survives a version step of undocumented lineage, whether the latent semantic readout survives, or whether directions derived from the lens keep their causal effect on the successor. That is the operational question here: when a model line is updated, must its interpretability instruments be refitted? The answer matters for monitoring systems that compare models over time. If a fitted readout is unstable across version updates, apparent changes in the readout may reflect instrument mismatch rather than changes in the model.

We study one version step within the Qwen 27B model line. The published Jacobian lens for Qwen3.6-27B [8] is applied without modification to Qwen3.8-27B [13]. Qwen3.8-27B was released less than four months (113 days) after Qwen3.6-27B [12, 13]; the checkpoints have matching architecture and tokenizer metadata. The vendor does not document whether the successor was produced by continued training of the earlier checkpoint, so the experiment is a test of functional transfer rather than a reconstruction of the training history.

The primary evaluation measures whether the transferred lens can retrieve an unstated intermediate entity in two-hop prompts. A second evaluation measures teacher-forced next-token rank on WikiText. We compare both readouts against the same lens on its home checkpoint and against an untransported logit-lens baseline. We also

report three supporting checks: a causal intervention derived from the home lens, a small sampling check on the home model, and an exploratory held-content example on the successor.

A separate exploratory follow-up asks whether the central two-hop result appears with a different lens artifact. A Jacobian lens fitted on Qwen3.6-35B-A3B is applied unchanged to Qwen3.5-35B-A3B Instruct and to a HauhauCS Qwen3.6-35B-A3B abliterated derivative using the same 40-prompt battery. Only a Q8_K_P build of the HauhauCS target was available; its surface-token check also uses a different protocol, and the causal intervention is not repeated. We therefore report the 35B measurements as a provisional lead for a matched replication, not as a replication of the 27B study. Together, the measurements ask how much of a fitted readout remains operational after related model changes; they do not by themselves identify which internal changes produced any observed difference.

2 Methods

2.1 Models, lens, and terminology

Qwen3.6-27B and Qwen3.8-27B each have 64 layers, hidden dimension d_{model}=5120, 24 attention heads, 4 key-value heads, and a 248,320-token vocabulary. Both models are evaluated in bf16. We refer to Qwen3.6-27B as the *home model* and Qwen3.8-27B as the *successor model*. The published Qwen3.6-27B Jacobian lens is the *home lens*; its unchanged application to Qwen3.8-27B is the *transferred lens*.

For layer ℓ, the lens contains a transport

$$J_\ell = \mathbb{E}\left[\frac{\partial h_{\text{final}}}{\partial h_\ell}\right]$$

estimated from the home model's local geometry. We use *Jacobian readout* for vocabulary logits computed after this transport and *raw readout* for logits computed by applying the unembedding directly to the intermediate state. These names distinguish the measured readouts from the global-workspace interpretation considered later. Figure 1 summarizes the readout and the transfer condition.

2.2 Readout measure and comparisons

At token position p, the Jacobian readout is unembed$(J_\ell h_{\ell,p})$ and the raw readout is unembed$(h_{\ell,p})$. The outcome is the rank assigned to a target token among all 248,320 vocabulary items; rank 1 is best. We evaluate layers $\{4, 8, \ldots, 60, 62\}$. One script, prompt set, position set, and ranking rule are used for all four combinations of model and readout.

Headline tables report the median target rank. Where paired tests are reported, comparisons are made at the prompt or token-position level. Sign tests exclude ties (the "decided" cases), and confidence intervals summarize the paired $\log_{10}$ rank ratio. These comparisons measure the combined effect of version change and home-lens mismatch; the design cannot separate those components.

2.3 Primary and secondary evaluation tasks

The primary task contains 40 English two-hop prompts. Each prompt names a property of a country and asks for another property while leaving the country itself unstated. For example, in "*Fact: The currency used in the country shaped like a boot is*," the target is the first token of *Italy*. The rank is measured at the final prompt position. The set contains a limited entity pool and repeats some targets; results therefore describe this prompt set rather than 40 independent entities.

The secondary task uses WikiText-103 test passages [7]. The first 100 passages longer than 800 characters are tokenized and truncated to 128 tokens. The teacher-forced next token is ranked at positions 32, 48, 64, 80, 96, 112, and 126, yielding 700 position samples per model. The task measures surface next-token readability under the same protocol on both checkpoints. It is not directly comparable with the published artifact-card evaluation, which uses a different protocol.

2.4 Exploratory 35B follow-up

The exploratory follow-up uses a second fitted artifact, the Qwen3.6-35B-A3B Jacobian lens released by stanleytheli [14]. It contains transports from source layers 0–38 into layer 39 and was fitted on 1,000 WikiText-103 prompts. We apply it unchanged to two compatible 40-layer mixture-of-experts targets: Qwen3.5-35B-A3B Instruct in BF16 [11], and the HauhauCS Qwen3.6-35B-A3B abliterated derivative in Q8_K_P [4]. The target model's own output norm and unembedding are used in each condition. As a decoder-identity gate, decoding the captured final-layer state reproduces the model's next-token argmax at every tested position on both targets. Because only the quantized HauhauCS build was available and the follow-up does not match the full primary protocol, its results are treated as provisional.

The latent-entity evaluation reuses the exact 40 prompts and first-token targets from the primary 27B task. Jacobian and raw ranks are computed at the last prompt position for all source layers. A separate surface-token check ranks the true next token at 212 positions from one fixed encyclopedic paragraph. Because that paragraph is not the 700-position WikiText battery and its home-model comparator comes from the artifact documentation, it is reported only as an order-of-magnitude check. No matched intervention was run for the 35B targets.

2.5 Supporting checks

The causal check uses directions computed entirely from the home lens. For token forms corresponding to *paradox*, *paradoxical*, and two Chinese equivalents, the pullback directions $J_\ell^\top u_w$ are orthogonalized within each layer and projected out of the residual stream at layers 18–47 during generation and activation capture. The prompt is "Describe Escher's impossible staircase." We cross model, clean versus ablated generation, and the model's reasoning mode enabled versus disabled, for eight cells in

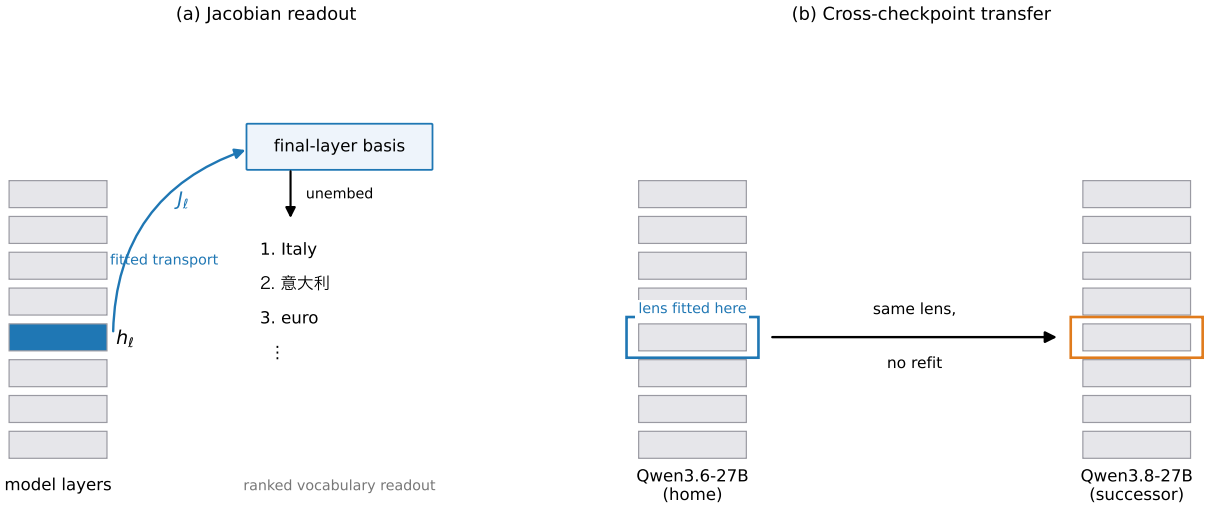


Figure 1: The instrument and the experiment. *(a)* The Jacobian readout: a fitted per-layer transport J_ℓ carries an intermediate state h_ℓ into the final-layer basis, where the model's own unembedding yields a ranked vocabulary readout (shown: the two-hop *Italy* example). *(b)* The transfer condition tested here: the lens fitted on Qwen3.6-27B is applied to Qwen3.8-27B without refitting.

total. This is a one-prompt, one-concept demonstration, not a general estimate of steering reliability.

Two further checks are exploratory. First, two behavioral prompt pairs on the home model are resampled at temperature 0.7 and compared with greedy captures. Second, a fixed self-description prompt with a forced completion prefix, previously characterized on the home model, is run once on the successor with the transferred lens at layers 39–55. These checks are reported separately from the two quantitative tasks and are not pooled with their estimates.

3 Results

We first report the main model-by-readout comparisons for the two-hop and WikiText tasks. The intervention, sampling, and held-content examples are then presented as supporting analyses.

3.1 Latent-entity readout

Figure 3 (left) shows that the home and transferred Jacobian readouts follow a similar depth profile. Across layers 24–48, where the target entity is most readable, the transferred lens assigns median ranks from 4 to 470. The raw readout assigns median ranks of roughly 10^3–10^4 on both models through the same band (Table 1).

The paired comparisons show that transfer is not uniform across depth. At layers 44–48, the transferred lens has a higher target rank than the home lens: median paired rank inflation is 1.7× at layer 44 and 3.4× at layer 48 (sign test, $p < 10^{-3}$; bootstrap 90% confidence intervals for the median $\log_{10}$ ratio are $[+0.15, +0.38]$ and $[+0.30, +0.65]$). Layer-40 readout is at parity ($p = 0.21$). At layers 24–36, the direction reverses. At layer 24, for example, the successor's median rank is 38 compared with 121 on the home model; 32 of 39 non-tied prompts favor the successor ($p < 10^{-3}$), with median ratios of 0.4–0.7× across this range.

This earlier-depth advantage has more than one possible source. It is consistent with the successor making the target content readable earlier, but the transfer cell changes both the underlying model and the match between model and lens. The experiment therefore cannot attribute the difference to one source. As a qualitative example, the transferred lens ranks *Italy* first at layer 40 on Qwen3.8-27B, with Chinese-script forms of *Italy* and *euro* also in the top five. Figure 2 shows the full depth-by-position readout for this prompt on both models.

3.2 Surface next-token readout

The WikiText comparison shows a smaller but consistent transfer penalty (Figure 3, right). At mid-network layers, paired per-position rank inflation is 1.2–1.3× (layer 32: $p = 3 \times 10^{-5}$). By layer 48 it is approximately 2× ($p < 10^{-30}$, n=700). The raw readout shifts by a similar factor at mid-network layers (1.2–1.3× at layers 24–40), so the mid-network inflation is not specific to the lens; the lens-specific cost is concentrated at layer 48, where the Jacobian readout inflates 2.0× against 1.2× for the raw readout (Table 1).

Thus the two tasks show different transfer profiles: latent-entity readout remains close to the home condition over its strongest band, whereas next-token readout shows a clearer late-layer cost. One interpretation is that version training changed directions involved in fine-grained token prediction more than those used by the latent-entity readout. Because the experiment does not isolate lens mismatch from model change, this remains an interpretation rather than a direct measurement of the training update.

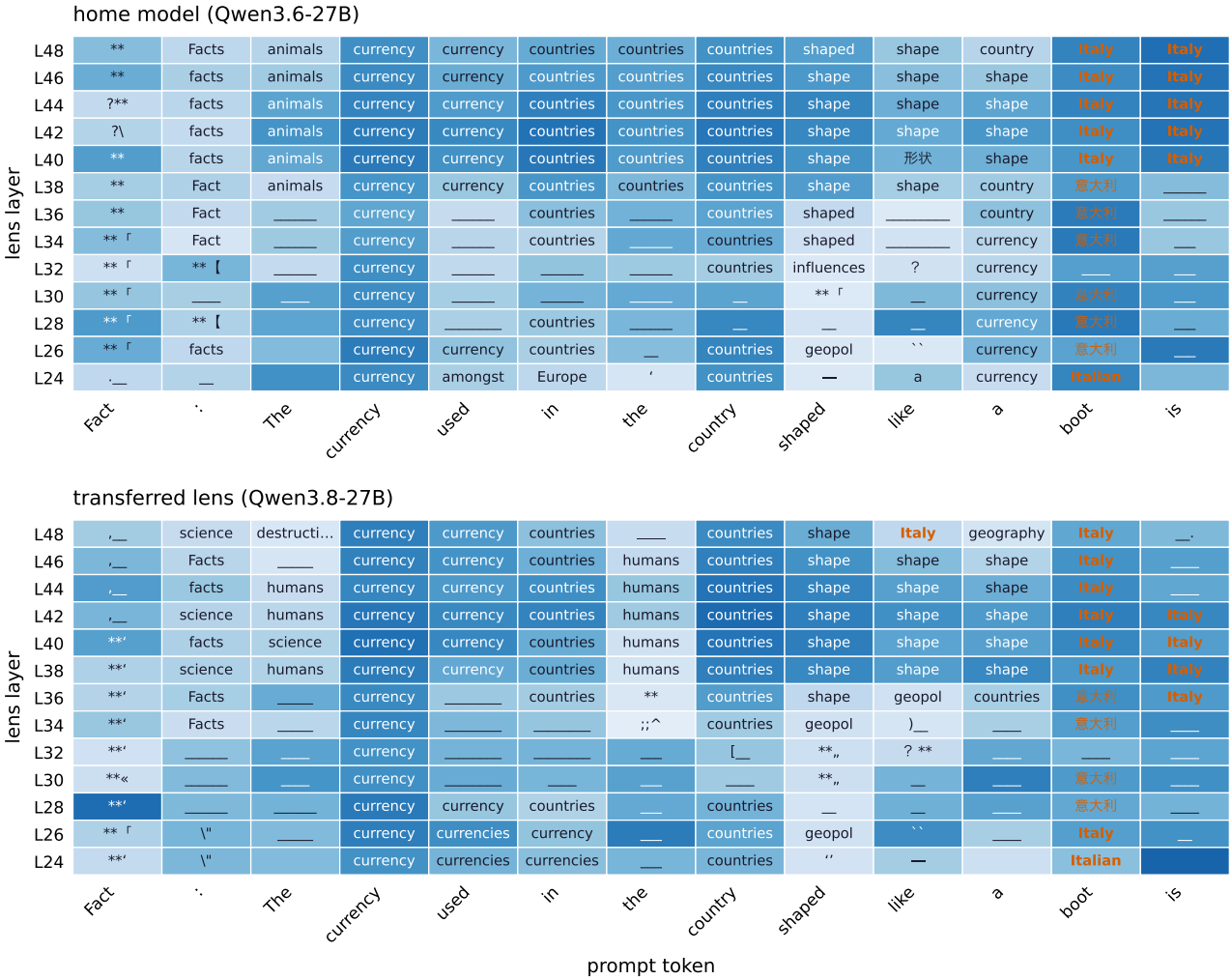


Figure 2: Transferred readout on one two-hop prompt, in the style of the workspace paper [3]. Each cell shows the top-ranked vocabulary token of the Jacobian readout at that layer and prompt position (prefill only); shading gives its probability mass, and orange marks the target family (*Italy*, *Italian*, Chinese-script *Italy*). The unstated middle entity becomes the leading readout well before the final position on both the home model (top) and the successor under the transferred lens (bottom). The capture is released with the dataset.

3.3 Exploratory 35B latent-entity result

The central latent-entity pattern also appears in the exploratory 35B measurements (Table 2). At the six summary layers from 20 to 38, the transferred Jacobian readout keeps the median target at or below rank 15.5 on both targets. On Qwen3.5-35B-A3B Instruct, the median best rank across the stack is 3 and 14 of 40 targets reach rank 1 somewhere in the stack. On the HauhauCS derivative, the corresponding values are 2 and 18 of 40. The raw readout is three to four orders of magnitude worse at layers 20–28 and catches up only near the output.

The separate 212-position surface-token check is also of the same order as the lens artifact's home-model documentation at layers 8, 16, 24, and 32, but the corpus and position protocol are not matched. We therefore use it as a calibration check rather than evidence for a quantitative transfer penalty. These provisional measurements show a similar latent-entity pattern with a second fitted lens and two related targets. They motivate a fully matched 35B replication but do not establish one.

3.4 Supporting analyses

Single-concept intervention. In all four clean cells, the generated descriptions contain the word *paradox*, and the lens assigns the paradox token group a pooled band mass of 124–157. In all four ablated cells, the word is absent and the corresponding readout is zero. The descriptions remain coherent and preserve the same central details, including the lithograph, closed loop, and illusion. Neighboring concepts such as *illusion* and *impossible* decrease two- to three-fold in the captured field but remain present in the text. Within this prompt and concept, directions computed from the Qwen3.6-27B lens therefore affect both the Qwen3.8-27B readout and its generated wording. The result supports transfer of this intervention but does not estimate how often such transfer will hold for other concepts or prompts.

Sampling check on the home model. At temperature 0.7, the surface wording changes across the two behavioral prompt pairs, while the leading never-generated words in the deep-band readout (words that rank highly

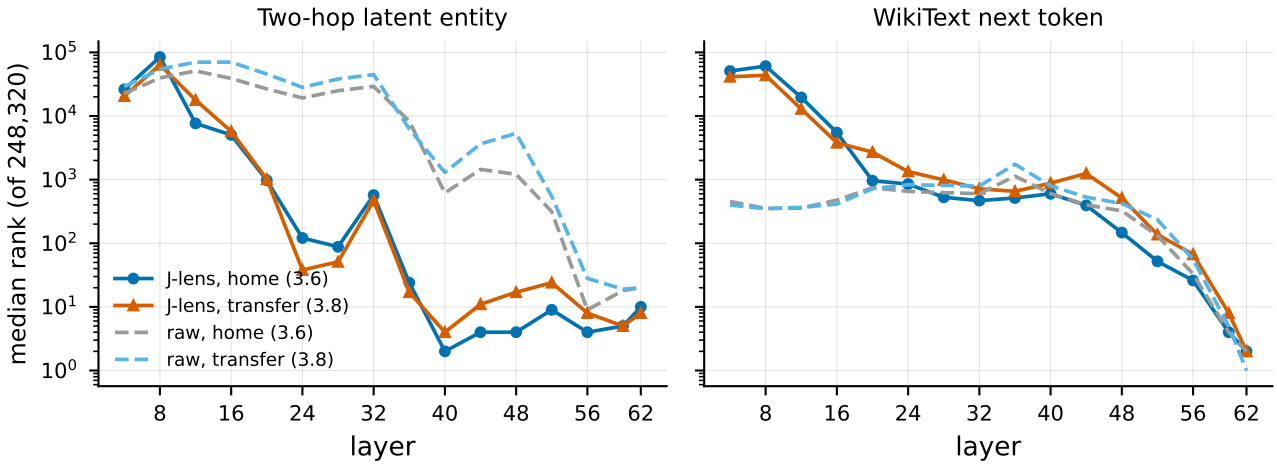


Figure 3: Median target-token rank by evaluation layer on a log scale. Lower is better. *Left*: the unstated intermediate entity in 40 two-hop prompts. *Right*: the teacher-forced next token at 700 WikiText positions. "Home" applies the Qwen3.6-27B lens to Qwen3.6-27B; "transfer" applies the same lens without modification to Qwen3.8-27B. "Raw" applies the unembedding without Jacobian transport.

	Two-hop latent entity				WikiText next token			
	home model		successor model		home model		successor model	
layer	J-lens	raw	J-lens	raw	J-lens	raw	J-lens	raw
8	84,539	39,620	63,868	54,560	60,810	353	43,645	349
16	5,042	38,980	5,739	70,285	5,486	477	3,798	413
24	121	19,219	38	28,224	852	654	1,339	832
32	570	29,185	470	44,822	468	601	719	791
40	2	617	4	1,299	600	603	881	802
48	4	1,213	17	5,341	147	323	514	421
56	4	9	8	28	26	33	67	56
62	10	20	8	20	2	2	2	1

Table 1: Median target-token rank for selected layers. The two-hop task uses 40 prompts at the final prompt position; the WikiText task uses 700 teacher-forced position samples per model. J-lens denotes the Jacobian readout. Raw denotes direct unembedding of the intermediate state.

	Qwen3.5-35B Instruct		HauhauCS Qwen3.6-35B	
layer	J-lens	raw	J-lens	raw
20	6.5	3,415	4.5	2,550
24	14	5,934	3	1,598
28	15.5	9,699	14	5,614
32	5.5	52.5	4	27.5
36	11.5	23	9	19
38	8	28.5	7	18.5

Table 2: Exploratory 35B follow-up on the same 40 two-hop prompts. Entries are median first-token target ranks (lower is better) for the Qwen3.6-35B-A3B lens applied unchanged. The protocol reports the latent-entity evaluation only.

in the readout but do not appear in the generated text) retain their identities and nearly the same ordering. Per-word mass varies by approximately a factor of two. In the verbatim-copy control, the sampled output is token-identical to the greedy output, and the capture also matches. This small check is consistent with local sampling stability, but it contains only two prompt pairs and does not test the successor model.

Exploratory held-content example. In the fixed self-description prompt, affect terms such as *feels* and *feelings* are among the strongest words read at layers 39–55 but do not appear in the generated output. The same qualitative signature had been observed on the home model. This example suggests that a previously observed form of latent–verbal divergence can also be found with the transferred lens. Because it is a single fixed prompt selected from earlier work, it is an illustration rather than an estimate of prevalence. The released capture contains the full field.

4 Discussion

4.1 What transfers in this experiment

The primary result is functional: a Jacobian lens fitted to Qwen3.6-27B continues to recover the target entity when applied unchanged to Qwen3.8-27B. Over the layers where the entity is most readable, the transfer cost is bounded to small rank factors and the readout remains much stronger than the raw baseline. The WikiText task shows that transfer is not cost-free; its penalty grows near the output end of the network. The intervention provides additional, narrower evidence that at least one direction derived from the home lens remains behaviorally relevant on the successor.

The exploratory 35B follow-up yields a similar latent-

entity pattern with a second fitted artifact and a mixture-of-experts architecture. Because one target is available only after Q8_K_P quantization, and because the follow-up does not match the WikiText protocol or repeat the intervention, it is a lead for replication rather than independent corroboration of the primary study.

These findings are compatible with partial stability of the verbalizable representational basis proposed by Gurnee et al. [3]. They are not a direct measurement of basis conservation. Similar readout performance can arise despite internal changes, and both fitted artifacts are tested only within related Qwen checkpoints or derivatives.

4.2 Implications for monitoring

Cross-checkpoint transfer can be treated as a measurable property rather than an assumption. For the matched 27B version step, the existing lens remains useful enough to compare selected readouts without first fitting a replacement. The provisional 35B result makes broader testing across fitted artifacts worthwhile but does not establish generality. Repeating the same transfer test at later releases would reveal when an instrument begins to drift or fail. The results therefore support the possibility of amortizing lens-fitting costs within a model line, but they are not sufficient to establish a once-per-line fitting rule.

4.3 Alternative explanations

The matched architecture and tokenizer may be important conditions for the observed transfer. The undocumented training relationship may also range from continued training of a shared base to a more complicated reuse of weights or data. In addition, the successor's better ranks at some middle layers cannot be assigned uniquely to earlier content resolution, because both model state and lens fit differ between the home and transfer conditions. A design with newly fitted lenses on both checkpoints, or additional intermediate releases, would help separate these explanations.

The 35B targets add further confounds: one is an older Instruct checkpoint, whereas the other is an abliterated derivative evaluated after Q8_K_P quantization. Their comparison does not isolate version, post-training, or quantization effects. The exploratory result is therefore evidence of operational readout under those combined changes, not an estimate of any one change's effect.

5 Related Work

Vocabulary-space readouts. The Jacobian lens was introduced by Gurnee et al. [3], with reference code and fitted lenses released for several open models [8]. Community-fitted artifacts now include a 397B-parameter mixture-of-experts model [10]. Wu et al. [15] subsequently use J-space as a recurrent interface for chain-of-thought. These studies motivate the practical question addressed here: whether an already fitted Jacobian lens remains usable after its model line is updated.

Comparing checkpoints. Model diffing commonly trains a shared object, such as a crosscoder, across two checkpoints to expose their differences [6]. The present test asks a complementary and narrower question. It does not train a representation across the pair; it measures how an existing instrument's readout changes when moved from one checkpoint to another.

Latent and verbalized content. Chain-of-thought can be an incomplete or unreliable account of the computation accompanying an answer [5, 1]. Vocabulary-space readouts offer a different view of intermediate content. Their value for monitoring successive model versions depends in part on whether changes in the readout can be distinguished from failures of the instrument itself, which is the stability question tested here.

6 Limitations

The study has six main boundaries. First, it tests one interpretability method, one fitted artifact in a matched transfer study, and a second artifact in an exploratory follow-up. Every source and target remains within the Qwen family and each transfer uses a compatible architecture and tokenizer. It does not establish transfer across model families, incompatible architectures, or larger version gaps.

Second, the same 40 English two-hop prompts are reused across the 27B and 35B tests. They draw from a limited entity pool with repeated targets and score only the target entity's first token. The 700 WikiText observations in the primary study come from fixed positions within 100 passages and should not be treated as 700 independent documents.

Third, the 35B follow-up does not match the full primary protocol. Its surface-token check uses one 212-position paragraph, one target is quantized and abliterated, and no matched causal intervention was run. The follow-up therefore motivates replication of the latent-entity result but does not itself support a replication claim.

Fourth, the main runs are greedy and single-seed. The temperature check covers only two prompt pairs on the 27B home model, so it cannot establish general sampling stability. The held-content result is likewise a selected, single-prompt qualitative example. Fifth, the causal check uses one concept, one prompt, and one intervention band, with no dose-response curve or broader prompt set. It demonstrates an effect in those cells but does not bound the generality or specificity of the intervention.

Finally, the transfer condition changes the model and the lens–model match at the same time. Paired tests localize and bound their combined effect, but they do not determine how much is caused by representational change in the successor and how much is caused by lens misfit. Any claim that the underlying workspace basis itself was conserved is therefore an inference from readout behavior, not a separately identified result.

7 Conclusion

This study answers its operational research question with a qualified no: a fitted lens need not always be refitted after a nearby, architecture-compatible model update. Applied unchanged to Qwen3.8-27B, the Qwen3.6-27B lens retained the unstated-entity signal (median rank 4 at home, 17 after transfer, and 5,341 in the successor's raw readout), and its *paradox* directions still altered both readout and generation. Yet the layer-dependent WikiText errors show that useful transfer did not preserve the lens's original calibration.

Cross-checkpoint reuse is therefore an empirical calibration problem, not a choice between assuming portability and refitting by default. For longitudinal interpretability and alignment monitoring, a lens can be carried forward when validation at the relevant layers, tasks, and targets shows that it remains fit for purpose. Measuring transfer error enables reuse while guarding against mistaking lens–model mismatch for change in the model. The decisive next step is to fit a Qwen3.8-27B lens, compare old-on-old, old-on-new, and new-on-new readouts, and expand the intervention across releases and concepts. That would separate model drift from instrument error and support an empirical threshold for refitting.

Reproducibility

Evaluation code, the two-hop prompt set, per-layer rank tables for all four model-by-readout cells, and the exploratory capture are available at https://huggingface.co/datasets/ec75hash/jacobian-lens-transfer-qwen36-38. The exploratory 35B follow-up is in the same dataset under `exploratory_35b/`: the two per-target S1/S2 result files, the exact prompt set, decoding and capture code, run logs, and a provenance note with the recorded model and lens hash prefixes.

AI Collaboration Statement

This paper was prepared with AI assistance. Claude (Anthropic) drafted evaluation code and earlier prose; OpenAI Codex assisted with manuscript revision and typesetting under the author's direction. All experiments, claims, and final text were reviewed and are the responsibility of the human author. Every reference was verified against its source.

References

[1] Iván Arcuschin, Jett Janiak, Robert Krzyzanowski, Senthooran Rajamanoharan, Neel Nanda, and Arthur Conmy. Chain-of-thought reasoning in the wild is not always faithful. arXiv:2503.08679, 2025. URL https://arxiv.org/abs/2503.08679.

[2] Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. arXiv:2303.08112, 2023. URL https://arxiv.org/abs/2303.08112.

[3] Wes Gurnee, Nicholas Sofroniew, Jack Lindsey, et al. Verbalizable representations form a global workspace in language models. arXiv:2607.15495, 2026. URL https://arxiv.org/abs/2607.15495. Also published in the Transformer Circuits Thread; reference implementation: https://github.com/anthropics/jacobian-lens.

[4] HauhauCS. Qwen3.6-35B-A3B-Uncensored-HauhauCS-Aggressive. Hugging Face model repository, 2026. URL https://huggingface.co/HauhauCS/Qwen3.6-35B-A3B-Uncensored-HauhauCS-Aggressive.

[5] Tamera Lanham, Anna Chen, Ansh Radhakrishnan, et al. Measuring faithfulness in chain-of-thought reasoning. arXiv:2307.13702, 2023. URL https://arxiv.org/abs/2307.13702.

[6] Jack Lindsey, Adly Templeton, Jonathan Marcus, Thomas Conerly, Joshua Batson, and Christopher Olah. Sparse crosscoders for cross-layer features and model diffing. Transformer Circuits Thread, 2024. URL https://transformer-circuits.pub/2024/crosscoders/index.html.

[7] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. arXiv:1609.07843, 2016. URL https://arxiv.org/abs/1609.07843.

[8] Neuronpedia. Jacobian lens artifacts for the qwen model family (revision qwen-n1000). Hugging Face model repository, 2026. URL https://huggingface.co/neuronpedia/jacobian-lens. Artifact file: qwen3.6-27b/jlens/Salesforce-wikitext/Qwen3.6-27B_jacobian_lens_n1000.pt.

[9] nostalgebraist. interpreting GPT: the logit lens. LessWrong, 2020. URL https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens.

[10] praxagent-org. Jacobian lens for Qwen3.5-397B-A17B. Hugging Face model repository, 2026. URL https://huggingface.co/praxagent-org/jacobian-lens-qwen3.5-397b-a17b.

[11] Qwen Team. Qwen3.5-35b-a3b model card. Hugging Face model repository, 2026. URL https://huggingface.co/Qwen/Qwen3.5-35B-A3B.

[12] Qwen Team. Qwen3.6-27b model card. Hugging Face model repository, 2026. URL https://huggingface.co/Qwen/Qwen3.6-27B.

[13] Qwen Team. Qwen3.8-27b model card. Hugging Face model repository, 2026. URL https://huggingface.co/Qwen/Qwen3.8-27B.

[14] stanleytheli. Jacobian lens for Qwen3.6-35B-A3B. Hugging Face model repository, 2026.

URL https://huggingface.co/stanleytheli/qwen3.6-35B-A3B-jlens. Artifact file: `lens.pt`; 1,000-prompt WikiText-103 fit.

[15] Junde Wu, Jiayuan Zhu, Fengling Liu, Minhao Hu, and Jiazhen Pan. J-CoT: Chain-of-thought in J-space. arXiv:2607.21981, 2026. URL https://arxiv.org/abs/2607.21981.